

General interpolation-supplemented lattice Boltzmann solution of irregular geometry neutron transport solving

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ABSTRACT

This study presents a general interpolation-supplemented lattice Boltzmann model (GILBM) for simulating neutron transport in complex geometries using non-uniform and non-orthogonal meshes. The physical domain with irregular geometry is transformed into a regular computational domain using the body-fitted mesh method, where the lattice Boltzmann method is applied. An additional interpolation step between migration and collision processes is used, maximizing the preservation of locality and parallelism inherent in the standard lattice Boltzmann method. The accuracy and flexibility of the GILBM are demonstrated through simulations of three representative neutron transport benchmark problems, including the neutron shielding problem, the heterogeneous 3×3 fuel pins problem, and the plate fuel problem. Numerical results exhibit good agreement with reference solutions, validating the effectiveness of the proposed methodology. This work can provide valuable insights into simulating neutron transport in complex geometries, contributing to developing more accurate and efficient simulation tools.

1. Introduction

Deterministic neutron transport simulation is a crucial component in nuclear reactor physics, which plays a predictive and guiding role in the application of nuclear reactor engineering. Due to its high-dimensional nature, its numerical discretization is typically carried out in a staged approach (Sanchez and McCormick, 1982). Firstly, the energy dimension is discretized using the multi-group approximation, which divides the continuous energy domain into a series of non-overlapping energy regions. By using the multi-group approximation, the multi-group neutron transport equation can be obtained. Subsequently, the angular domain is treated using basis function expansions, such as the spherical harmonic (P_N) method (Cao and Wu, 2004; Fletcher, 1983; Reed, 1972), simplified P_N method (Duerigen, 2013), wavelet expansion method (Cao et al., 2008; Chi et al., 2023), or discretization methods like the discrete-ordinate (S_N) method (Keller, 1960; Kong et al., 2024; Yang et al., 2022). Through this angular treatment, the high-dimensional equation is transformed into a set of coupled partial differential equations (PDEs). Various temporal-spatial discretization methods can be employed to solve these equations, including detailed methods such as the method of characteristics (Boyd et al., 2014; Chen et al., 2008; Deo et al., 2014;

Hoffman and Lee, 2016; Kochunas, 2013; Mazumdar and Degweker, 2015; Wang et al., 2020b), finite-volume method (Hu et al., 2017; Ma et al., 2021), finite-element method (Briggs et al., 1975; Chan and Xiao, 2021; Fang et al., 2020; Guo et al., 2021; Lee and Joo, 2014), finite-differential method (Morató et al., 2020), meshfree method (Kashi et al., 2017), and coarse-mesh method such as the variational nodal method (Sun et al., 2024; Yin et al., 2024; Zhang et al., 2018), and analytical nodal method (Rocheleau and Wang, 2020).

The lattice Boltzmann method (LBM) is a novel mesoscopic computational fluid dynamics approach (Han and Cundall, 2013; Kuznik et al., 2010; Zhang et al., 2021; Zhao et al., 2010), that can be employed as an alternative technique for detailed neutron transport calculations (Bindra and Patil, 2012; Wang et al., 2017b). By simulating physical processes through the mesoscopic collision and migration of distribution functions, the LBM offers a unique perspective. Its strong localization yields several advantages, including ease of implementation, simplicity, and inherent parallelism. These advantages have led to the extension of the LBM to neutron/photon transport problems (Asinari et al., 2010; Bindra and Patil, 2012; Gairola and Bindra, 2017; Ma et al., 2019; Mishra and Vernekar, 2012; Vernekar and Mishra, 2014; Wang et al., 2023; Wang et al., 2017b). Although the standard LBM (SLBM) has many advantages

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and extensive applications, some limitations still need to be improved, such as the requirement for uniform orthogonal meshes and high computational costs (Yang et al., 2017), which restrict its applicability in solving practical engineering problems with complex geometries, such as the circular pins in standard light water reactors (LWRs).

When employing uniform square meshes to approximate irregular geometries, dense meshing is often required, which may substantially increase computational resource demands. To improve these conditions, several advanced techniques can be utilized. The first idea is to use non-uniform meshes, including multi-block method (Farhat et al., 2010; Liu et al., 2009; Yu and Girimaji, 2006) and adaptive-mesh-refinement method (Huang and Wu, 2016; Misaka et al., 2018; Usui et al., 2017), as well as multi-block adaptive-mesh-refinement method (Ma et al., 2019; Wang et al., 2020a; Wang et al., 2018) that combines multi-block method and adaptive-mesh-refinement method. This strategy enables local mesh densification in complex geometries while using coarser meshes in less intricate regions, thereby achieving a balance between accuracy and computational complexity. However, the implementation of this approach can be challenging. An alternative idea is to employ non-standard LBM, such as the finite-difference LBM (Reider and Sterling, 1995; Wang et al., 2017a; Yang et al., 2017), the finite-volume LBM (Li and Li, 2018; Peng et al., 1999; Stiebler et al., 2006) and the finite-element LBM (Alekseenko, 2011; Matin et al., 2018; Wu et al., 2018). These methods discretize the discrete velocity Boltzmann equation using classical spatial discretization techniques, allowing for the use of different mesh structures, including non-uniform and unstructured meshes. Nevertheless, the advantages of the standard LBM, namely its simplicity and strong parallelism, rely on the specific differential format of the first-order upwind discretization of space and time derivatives. When using non-standard LBMs, this special discretization is no longer maintained, potentially compromising the simplicity and parallelism of the standard LBM (this will be further discussed in Section 2.1).

In comparison to existing methods, the interpolation-supplemented LBM (ISLBM) (He and Doolen, 1997) offers a more viable alternative. By introducing an interpolation step between collision and migration processes of SLBM, the ISLBM ensures accurate distribution function migration to neighboring mesh points on non-orthogonal grids, effectively mapping complex physical geometries to a standard orthogonal computational domain while preserving the collision-migration process of SLBM. This significantly reduces the difficulty of implementation. However, the ISLBM's reliance on analytical functions to map the physical domain to the computational domain limits its applicability in complex geometries that cannot be analytically described. To improve this application, researchers (Imamura et al., 2005b) proposed a generalized interpolation-supplemented lattice Boltzmann method (GILBM), which incorporates a body-fitted coordinate mesh into the LBM, enabling flexible application on arbitrary non-orthogonal meshes and overcoming the ISLBM's limitations in complex geometries. The GILBM's notable advantages have led to its widespread adoption in various fields, including gas dynamics (Reyes Barraza and Deiterding, 2020; Zhong et al., 2020), fluid flow (Imamura et al., 2005b; Li et al., 2015), heat transfer (Ren et al., 2024), and noise radiation (Song et al., 2024a; Song et al., 2024b), etc.

In this work, the GILBM is extended to neutron transport LBM simulation in irregular geometries. As industrial applications predominantly use established methods like the method of characteristics and collision probability techniques, which can deliver accurate results with relatively coarse meshes, the proposed GILBM approach is not intended to replace these mature industrial-grade numerical methods. Instead, the main purpose of this study is to improve the limitations of traditional LBM in irregular geometry, as a complementary computational tool for addressing strong geometric distortion problems, such as fuel rod irradiation, bending, and coolant channel deformation. The rest of this paper is structured as follows: in Section 2, the mathematical theories of the proposed GILBM for neutron transport are described, including the SLBM for neutron transport, the construction of body-fitted meshes, the

application of GILBM to neutron transport, and its extension to multi-mesh coupling problems via the use of different body-fitted meshes in distinct regions and their subsequent coupling; in Section 3, some typical benchmark problems are simulated to verify the accuracy and flexibility of GILBM; Section 4 draws some conclusions of this work.

2. General interpolation-supplemented lattice Boltzmann method for neutron transport

This section outlines the detailed establishment of the general interpolation supplement lattice Boltzmann method (GILBM) for solving the neutron transport equation on body-fitted meshes, including the standard LBM for neutron transport, the body-fitted GILBM, and the domain decomposition approach of GILBM.

2.1. Lattice Boltzmann method for neutron transport

To provide a comprehensive foundation, we commence by introducing the standard lattice Boltzmann method (SLBM) model for neutron transport. The two-dimensional discrete-ordinate neutron transport equation (NTE), which accounts for multi-group isotropic scattering and absorption in materials, can be expressed as:

$$\frac{1}{v_g} \frac{\partial \phi_{\alpha,g}(\mathbf{r}, t)}{\partial t} + \mu_{\alpha} \frac{\partial \phi_{\alpha,g}(\mathbf{r}, t)}{\partial x} + \eta_{\alpha} \frac{\partial \phi_{\alpha,g}(\mathbf{r}, t)}{\partial y} + \Sigma_{t,g}(\mathbf{r}) \phi_{\alpha,g}(\mathbf{r}, t) = \sum_{g'=1}^G \Sigma_{s,g' \rightarrow g}(\mathbf{r}) \sum_{\alpha'=1}^{N_{\alpha}} \Delta \Omega_{\alpha'} \phi_{\alpha',g'}(\mathbf{r}, t) + Q_g(\mathbf{r}, t), \quad (1)$$

where α and g represent the indexes of the angle discretization direction and energy group, respectively; v_g is the neutron speed of energy g ; N_{α} and G represent the total number of angle discretization directions and energy groups, respectively; μ_{α} and η_{α} represent the projections of the angle discretization direction onto x -axis and y -axis, respectively; $\Delta \Omega_{\alpha'}$ represents the weight coefficients corresponding to the α' -th angle discretization direction; $\phi_{\alpha,g}$ represents the g -th neutron angular flux along α -th angle discretization direction; $\mathbf{r}$ represents the spatial positioning and t represents the time; $\Sigma_{t,g}$ represents the g -th group macroscopic total cross-section, and $\Sigma_{s,g' \rightarrow g}$ represents the macroscopic scattering cross-section from g' -th energy group to g -th energy group; Q_g represents the neutron source, including the fission source and external source.

To use LBM, the NTE must be reformulated in a conservative form, which can be expressed as:

$$\partial_t \phi_{\alpha,g} + \nabla \cdot (\mathbf{u}_{\alpha,g} \phi_{\alpha,g}) = F_{\alpha,g} \quad (2)$$

where $\mathbf{u}_{\alpha,g}$ is the neutron speed vector defined as:

$$\mathbf{u}_{\alpha,g} = v_g [\mu_{\alpha}, \eta_{\alpha}]^T \quad (3)$$

and $F_{\alpha,g}$ is defined as the general neutron source as:

$$F_{\alpha,g} = v \left(\sum_{g'=1}^G \Sigma_{s,g' \rightarrow g} \sum_{\alpha'=1}^{N_{\alpha}} \Delta \Omega_{\alpha'} \phi_{\alpha',g'} + Q_g - \Sigma_{t,g} \phi_{\alpha,g} \right) \quad (4)$$

Subsequently, the LBM model can be formulated for the NTE in conservative form. To illustrate this process, we consider the mono-energy NTE as a representative example, and note that the multi-group case can be readily extended by introducing an additional subscript to denote energy group g and considering the multi-group neutron source. Different from the convection numerical discretization method, the LBM is a "bottom-up" modeling approach, whose basic principle relies on the discrete-velocity Boltzmann equation:

$$\frac{\partial f_{\alpha,i}(\mathbf{r}, t)}{\partial t} + \mathbf{c}_i \cdot \nabla f_{\alpha,i}(\mathbf{r}, t) = -\frac{1}{\tau_c} (f_{\alpha,i}(\mathbf{r}, t) - f_{\alpha,i}^{eq}(\mathbf{r}, t)) + F_{\alpha,i}(\mathbf{r}, t) \quad (5)$$

where $f_{\alpha,i}(\mathbf{r}, t)$ represents the local distribution functions corresponding

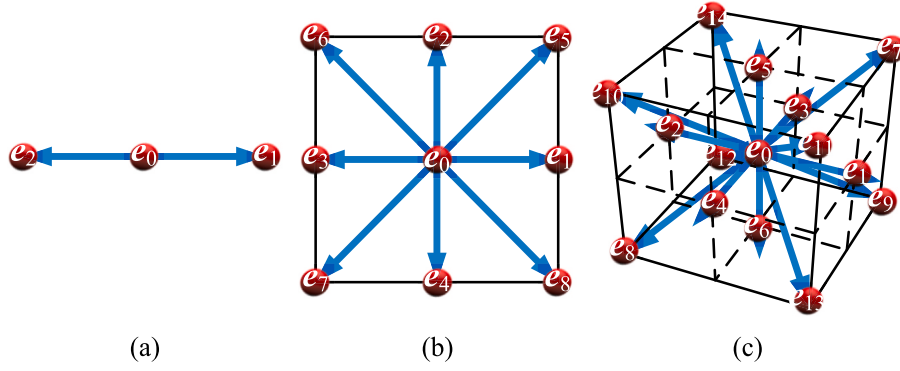


Fig. 1. Schematic of typical DdQq lattice speed models.

to angular direction α , lattice direction i , position $\mathbf{r}$ and time t ; $f_{a,i}^{eq}(\mathbf{r}, t)$ is so called the equilibrium distribution function of LBM theory; $F_{a,i}(\mathbf{r}, t)$ represents the lattice source term; τ_c is the relaxation time of LBM theory; $\mathbf{c} = [c_x, c_y]$ represents the lattice speed with c_x, c_y being the x and y -direction components. Typically, the DdQq (d -dimensional condition with q lattice speeds) lattice speed model is employed. The commonly used DdQq lattice speed models, including the D1Q3, D2Q9, and D3Q15 models, are illustrated in Fig. 1, and the corresponding lattice parameters are defined as:

D1Q3:

$$\mathbf{c} = (0, -1, 1)c \quad (6)$$

$$\varpi_0 = 2/3, \varpi_1 = \varpi_2 = 1/6, \quad c_s^2 = c^2/3$$

D2Q9:

$$\mathbf{c} = \begin{pmatrix} 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \end{pmatrix} c \quad (7)$$

$$\varpi_0 = 4/9, \varpi_{1-4} = 1/9, \varpi_{5-8} = 1/36, \quad c_s^2 = c^2/3$$

D3Q15:

$$\mathbf{c} = \begin{pmatrix} 0 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \end{pmatrix} c \quad (8)$$

$$\varpi_0 = 2/9, \varpi_{1-6} = 1/9, \varpi_{7-14} = 1/72, \quad c_s^2 = c^2/3$$

where ϖ, c_s represent the lattice weight function and lattice sound speed, respectively; $c = \Delta x/\Delta t$ is the lattice migration speed with Δx being the uniform mesh size, and Δt being the time step size. In the standard LBM, all meshes have the same size in all directions, i.e., $\Delta x = \Delta y = \Delta z$.

By using the first-order upwind discretization of time and space derivatives as:

$$\begin{aligned} & \frac{\partial f_{a,i}(\mathbf{r}, t)}{\partial t} + \mathbf{c}_i \cdot \nabla f_{a,i}(\mathbf{r}, t) \\ &= \frac{f_{a,i}(\mathbf{r} + \mathbf{c}_i \Delta t, t + \Delta t) - f_{a,i}(\mathbf{r} + \mathbf{c}_i \Delta t, t)}{\Delta t} + c \frac{f_{a,i}(\mathbf{r} + \mathbf{c}_i \Delta t, t) - f_{a,i}(\mathbf{r}, t)}{\Delta x} \quad (9) \\ &= \frac{f_{a,i}(\mathbf{r} + \mathbf{c}_i \Delta t, t + \Delta t) - f_{a,i}(\mathbf{r}, t)}{\Delta t} \end{aligned}$$

the simple lattice Boltzmann equation can be deduced as:

$$\begin{aligned} & f_{a,i}(\mathbf{r} + \mathbf{c}_i \Delta t, t + \Delta t) = f_{a,i}(\mathbf{r}, t) \\ & - \frac{1}{\tau} \left(f_{a,i}(\mathbf{r}, t) - f_{a,i}^{eq}(\mathbf{r}, t) \right) + \Delta t F_{a,i}(\mathbf{r}, t) + \frac{\Delta t^2}{2} \partial_t F_{a,i}(\mathbf{r}, t) \quad (10) \end{aligned}$$

where $\tau = \tau_c/\Delta t$ is the dimensionless relaxation time; The last term on the right-hand side of this equation constitutes a correction term. Notably, for steady-state simulation, Eq. (10) is essentially an iterative scheme, where the time step Δt is virtual and serves as a parameter

controlling the stability of iteration progression.

When solving different governing equations, the LBM equation (10) remains consistent, requiring only minor adjustments to the definitions of the equilibrium distribution function $f_{a,i}^{eq}(\mathbf{r}, t)$, source distribution function $F_{a,i}(\mathbf{r}, t)$, and dimensionless relaxation time τ . This is one of the prominent advantages of LBM in multi-physics coupling calculations, where a single solver can be applied to simulate different physical processes through minor modifications. To apply the LBM to solve the NTE, we need to employ the Chapman-Enskog multi-scale expansion to establish an equivalent correspondence between the LBM equation (10) and the NTE (2), thereby determining the undetermined variables in the LBM method. According to the related article (Wang et al., 2017b), these undetermined variables have been determined as:

$$f_{a,i}^{eq} = \varpi_i \phi_a \left[1 + \frac{\mathbf{c}_i \cdot \mathbf{u}_a}{c_s^2} + \frac{[\mathbf{u}_a \mathbf{u}_a - c_s^2 \mathbf{I}] : (\mathbf{c}_i \mathbf{c}_i - c_s^2 \mathbf{I})}{2c_s^4} \right] \quad (11)$$

$$F_a = \nu \left(\frac{\sum_s}{4\pi} \sum_{\alpha'} \phi_{\alpha'} \Delta \Omega_{\alpha'} + Q_a - \Sigma_i \phi_a \right) \quad (12)$$

$$F_{a,i} = \varpi_i F_a \left[1 + \frac{\mathbf{c}_i \cdot \mathbf{u}_a}{2c_s^2} \right] \quad (13)$$

$$\tau = 1 \quad (14)$$

where $\mathbf{I}$ represents the identity matrix; the operator “:” represents the double dot product between matrices; the neutron angular flux ϕ and scalar flux Φ can be calculated as:

$$\phi_a = \sum_i f_{a,i} = \sum_i f_{a,i}^{eq} \quad (15)$$

$$\Phi = \sum_{a=1}^{N_a} \phi_a \Delta \Omega_a \quad (16)$$

With a dimensionless relaxation time of $\tau = 1$, the LBM equation (10) can be simplified to a finite-Boltzmann format as:

$$f_{a,i}(\mathbf{r} + \mathbf{c}_i \Delta t, t + \Delta t) = f_{a,i}^{eq}(\mathbf{r}, t) + \Delta t F_{a,i}(\mathbf{r}, t) + \frac{\Delta t^2}{2} \partial_t F_{a,i}(\mathbf{r}, t) \quad (17)$$

The solution of this equation can be divided into collision and migration processes as:

Collision:

$$f_{a,i}^* (\mathbf{r}, t) = f_{a,i}^{eq} (\mathbf{r}, t) + \Delta t F_{a,i} (\mathbf{r}, t) + \frac{\Delta t^2}{2} \partial_t F_{a,i} (\mathbf{r}, t) \quad (18)$$

Migration:

$$f_{a,i}(\mathbf{r} + \mathbf{c}_i \Delta t, t + \Delta t) = f_{a,i}^* (\mathbf{r}, t) \quad (19)$$

where $f_{a,i}^*$ is the post-collision distribution function. Above is the brief

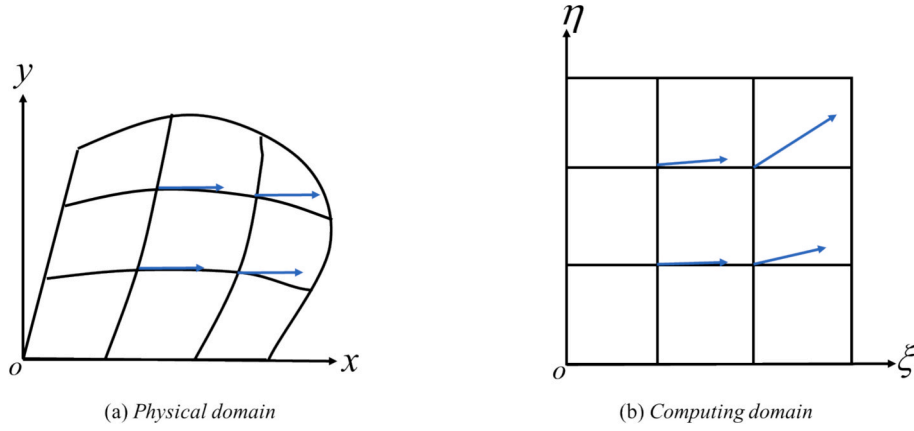


Fig. 2. Mapping relationship between (a) physical domain and (b) computing domain.

description of the neutron transport lattice Boltzmann model, in which the distribution function describes the evolution of neutron flux along the LBM lattice direction. For the multi-group conditions, the above LBM model for mono-energy neutron transport can be directly extended by modifying the source term F to a multi-group source term (4) and applying the LBM model sequentially to each energy group.

The collision step is totally localized, while the migration step is totally linear. Therefore, the lattice Boltzmann model shows a simple implementation and strong parallel features. However, its applicability is limited by the requirement of a uniform orthogonal mesh, which poses significant challenges in simulating complex geometries. To mitigate this limitation, non-standard LBM models, such as finite-element and finite-volume LBM, can be developed. Nevertheless, these models involve discretizing the spatial derivative of Eq. (5) using finite-element or finite-volume methods, which compromises the simplicity of the linear LBM Eq. (17) and loses the separation of collision and migration steps. As a result, the non-standard LBM models destroy the advantages of simple implementation and robust parallelization inherent to traditional LBM. Therefore, to enhance the adaptability of LBM to complex geometric conditions while maintaining the outstanding advantages of standard LBM, the following sections will describe the extension of SLBM and GILBM.

2.2. Body-fitted GILBM model

2.2.1. GILBM

The standard LBM requires a uniform square mesh configuration, wherein the distribution function coincides with adjacent mesh nodes after the migration process, thereby limiting its application to complex geometries. To overcome this limitation, the interpolation-supplemented lattice Boltzmann method (ISLBM) (He and Doolen, 1997) introduces an interpolation step between collision and migration steps, ensuring that the distribution functions after migration fall on mesh points. The GILBM (Imamura et al., 2005a) further extends ISLBM by eliminating the dependence on analytical expressions of geometric shapes.

The GILBM framework is established using body-fitted meshes, which transform irregular physical domains into regular (typically square) computational domains. This transformation involves establishing a mapping relationship between boundary points of the physical and computational domains, thereby relating mesh nodes in both domains. To guarantee numerical stability and reliability, the generated meshes must satisfy orthogonality, smoothness, and controllability of mesh density in appropriate body coordinates. To meet these requirements, this work employs the partial-differential equation method, which determines the physical coordinates corresponding to each node in the computational domain by solving a partial-differential equation.

Taking two-dimensional condition as an example, $\mathbf{r} = (x, y)$ and $\xi = (\xi, \eta)$ represent the physical domain in generalized coordinates and the computing domain in orthogonal coordinates, respectively, as shown in Fig. 2. The relationship between these two coordinates is given by:

$$\frac{\partial^2 \xi}{\partial x^2} + \frac{\partial^2 \xi}{\partial y^2} = P(\xi, \eta) \quad (20)$$

$$\frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2} = Q(\xi, \eta) \quad (21)$$

where P and Q are regulatory factors for adjusting mesh density. Commonly, direct solutions of Eqs. (20) and (21) are challenging to obtain. By applying the inverse transformation, the physical coordinate $\mathbf{r} = (x, y)$ can be derived from the computing coordinate $\xi = (\xi, \eta)$ as:

$$\alpha \frac{\partial^2 x}{\partial \xi^2} - 2\beta \frac{\partial^2 x}{\partial \xi \partial \eta} + \gamma \frac{\partial^2 x}{\partial \eta^2} + J^2 \left(\frac{\partial^2 x}{\partial \xi^2} P + \frac{\partial^2 x}{\partial \eta^2} Q \right) = 0 \quad (22)$$

$$\alpha \frac{\partial^2 y}{\partial \xi^2} - 2\beta \frac{\partial^2 y}{\partial \xi \partial \eta} + \gamma \frac{\partial^2 y}{\partial \eta^2} + J^2 \left(\frac{\partial^2 y}{\partial \xi^2} P + \frac{\partial^2 y}{\partial \eta^2} Q \right) = 0 \quad (23)$$

$$\alpha = \left(\frac{\partial x}{\partial \eta} \right)^2 + \left(\frac{\partial y}{\partial \eta} \right)^2, \quad \gamma = \left(\frac{\partial x}{\partial \xi} \right)^2 + \left(\frac{\partial y}{\partial \xi} \right)^2, \quad (24)$$

$$\beta = \frac{\partial x}{\partial \xi} \frac{\partial x}{\partial \eta} + \frac{\partial y}{\partial \xi} \frac{\partial y}{\partial \eta}, \quad J = \frac{\partial x}{\partial \xi} \frac{\partial y}{\partial \eta} - \frac{\partial y}{\partial \xi} \frac{\partial x}{\partial \eta}.$$

The Jacobian matrix J represents the area scaling factor between the physical and computational domains, i.e., the ratio of the area of a given mesh in the two domains. The coordinate transformation can be expressed in terms of J as:

$$\begin{bmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \xi}{\partial y} \\ \frac{\partial \eta}{\partial x} & \frac{\partial \eta}{\partial y} \end{bmatrix} = \frac{1}{J} \begin{bmatrix} \frac{\partial y}{\partial \eta} & -\frac{\partial x}{\partial \eta} \\ -\frac{\partial y}{\partial \xi} & \frac{\partial x}{\partial \xi} \end{bmatrix} \quad (25)$$

By employing the computational coordinates, the finite-Boltzmann-formatted neutron transport lattice Boltzmann equation can be formulated as:

$$f_{a,i}(\xi + \tilde{\mathbf{c}}_i \Delta t, t + \Delta t) = f_{a,i}^{eq}(\xi, t) + \Delta t F_{a,i}(\xi, t) + \frac{\Delta t^2}{2} \partial_t^2 F_{a,i}(\xi, t) \quad (26)$$

where $\tilde{\mathbf{c}}_i$ is the lattice speed in computing coordinates. Due to the localization of the collision step, Eq. (18) can be converted to the computing domain by replacing the position vector $\mathbf{r}$ with ξ as:

$$f_{a,i}^*(\xi, t) = f_{a,i}^{eq}(\xi, t) + \Delta t F_{a,i}(\xi, t) + \frac{\Delta t^2}{2} \partial_t^2 F_{a,i}(\xi, t) \quad (27)$$

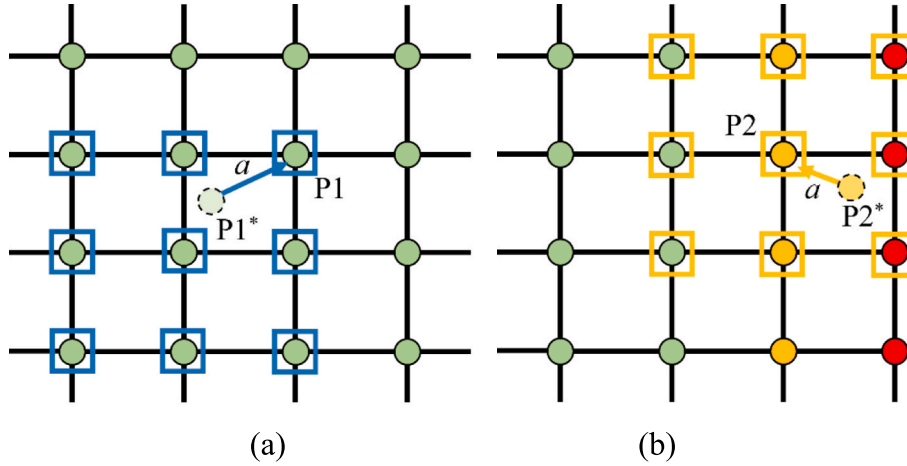


Fig. 3. Schematic of GILBM interpolation for (a) interior node and (b) near boundary node.

In contrast, for the migration step, although the migration direction and speed of all the physical nodes are consistent, as shown in Fig. 2(a), the equivalent velocities in the computational domain vary due to the coordinate transformation, as shown in Fig. 2(b). In considering this condition, the lattice speed $\tilde{c}_i$ is defined as:

$$c_\xi = \frac{\partial \xi}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial \xi}{\partial y} \frac{\partial y}{\partial t} = c_x \frac{\partial \xi}{\partial x} + c_y \frac{\partial \xi}{\partial y} \quad (28)$$

$$c_\eta = \frac{\partial \eta}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial \eta}{\partial y} \frac{\partial y}{\partial t} = c_x \frac{\partial \eta}{\partial x} + c_y \frac{\partial \eta}{\partial y} \quad (29)$$

The migration step of Eq. (19) is rewritten as:

$$f_{a,i}(\xi, t + \Delta t) = f_{a,i}^*(\xi - \Delta \xi, t) \quad (30)$$

where $\Delta \xi$ represents the migration distance in the computing domain. In the physical domain, since the lattice speed $\mathbf{c}$ is constant, the migration distance can be calculated directly by multiplying $\mathbf{c}$ and Δt . However, in the computing domain, the lattice speed varies at different positions, and the migration path may deviate from a straight line. Consequently, the migration distance in the computational domain should be computed using:

$$\Delta \xi = \int_0^{\Delta t} \tilde{\mathbf{c}} dt \quad (31)$$

By using the second-order Runge-Kutta scheme, the Eq. (31) can be solved using two steps:

$$\Delta \xi^{(1)} = \frac{1}{2} \tilde{\mathbf{c}} \Delta t \quad (32)$$

$$\Delta \xi = \Delta t \tilde{\mathbf{c}}(\xi - \Delta \xi^{(1)}) \quad (33)$$

By substituting Eqs. (32) and (33) into Eq. (30), it can be found that the migrated points may not necessarily fall on adjacent grid points, so an interpolation-supplemented method is needed to calculate the distribution functions before migration. Since the LBM is a second-order method due to its Lagrange characteristics (Chen and Doolen, 1998), the second-order interpolation is used. A schematic of GILBM interpolation is presented in Fig. 3, in which the green, red, and yellow circles denote interior nodes, boundary nodes, and near-boundary nodes, respectively. The blue and yellow hollow boxes represent interpolation nodes. For interior nodes, as shown in Fig. 3(a), the post-collision distribution function at node P1* can be obtained using a second-order upwind-interpolated scheme as:

$$f^*(\xi - \Delta \xi) = \sum_{m=0}^2 \sum_{n=0}^2 a_{m,1} a_{n,2} f^*(\xi + m i_d, \eta + n j_d) \quad (34)$$

here we omit the subscript α and i ; i_d and j_d represent the upwind direction. The interpolation coefficients $a_{m,1}$ and $a_{m,2}$ are defined as:

$$\begin{aligned} a_{0,k} &= \frac{1}{2} (|\Delta \xi_{up,k}| - 1) (|\Delta \xi_{up,k}| - 2), \\ a_{1,k} &= -|\Delta \xi_{up,k}| (|\Delta \xi_{up,k}| - 2), \\ a_{2,k} &= \frac{1}{2} |\Delta \xi_{up,k}| (|\Delta \xi_{up,k}| - 1), \end{aligned} \quad (35)$$

where the subscript k represents the coordinate component, for a two-dimensional condition, $k = 1, 2$; $\Delta \xi_{up,k}$ represents the upwind migration distance along the k -th coordinate.

For mesh nodes adjacent to the boundary, such as P2* in Fig. 3(b), the second-order upwind interpolation scheme in Eq. (34) becomes inapplicable due to the limited number of upstream nodes in the inward directions. Inside, a second-order central interpolation scheme is employed to obtain the values of the distribution functions:

$$f^*(\xi - \Delta \xi) = \sum_{m=-1}^1 \sum_{n=-1}^1 b_{m,1} b_{n,2} f^*(\xi + m i_d, \eta + n j_d) \quad (36)$$

$$\begin{aligned} b_{-1,k} &= \frac{1}{2} \Delta \xi_{up,k} (\Delta \xi_{up,k} - 1), \\ b_{0,k} &= 1 - \Delta \xi_{up,k}^2, \\ b_{1,k} &= \frac{1}{2} \Delta \xi_{up,k} (\Delta \xi_{up,k} + 1). \end{aligned} \quad (37)$$

For both interior and near-boundary nodes, the coefficients a and b depend solely on the grid mapping, allowing them to be pre-computed prior to the GILBM iteration. This pre-calculation step enables a reduction in the interpolation time associated with the GILBM. To ensure the numerical stability and control the distribution function to fall within a grid during each migration, the time step is defined as:

$$\Delta t = \lambda \min \left| \frac{1}{\tilde{\mathbf{c}}} \right| \quad (38)$$

where $0 < \lambda < 1$ is so called CFL number. In the computational domain, the mesh size is set to 1, and the selection of the time step considers all migration velocity components in all grid points of the computational domain, ensuring that the migration process at all positions and in all directions can be performed within one mesh range.

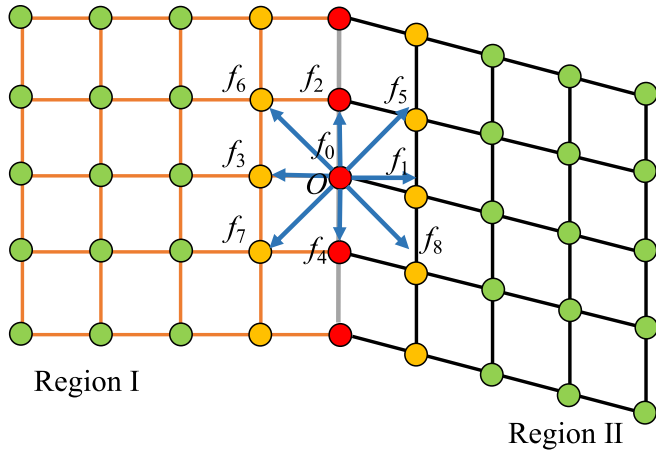


Fig. 4. Schematic of the physical domain with different body-fitted meshes connecting.

2.2.2. Processes of multiple body-fitted meshes

Although the GILBM with body-fitted meshes can be used to simulate various complicated geometries, for some practical conditions, using a partitioned body-fitted mesh will bring better results.

In practical applications, such as the heterogeneous fuel pins, a complete computational domain is always composed of sub-domains with different characteristics. To address these challenges, it is essential to employ separate body-fitted grids for distinct regions. At the interfaces between these grids, discontinuities and non-smooth transitions may occur, as the adjacent grids may not share a common computational domain, thereby underscoring the importance of specialized treatment methods for these interfacial regions.

Taking Fig. 4 as an example to illustrate the intersection of two body-fitted meshes, where orange and black lines denote distinct meshes, and gray lines indicate the interface between regions I and II. The grid points are categorized into three types: red (shared boundary points), yellow (near-boundary points), and green (internal points). For green and orange grid points, distribution functions in various migration directions can be accurately recovered using second-order upwind or central interpolation, thereby preserving macroscopic quantities.

For mesh nodes located at the intersection of computational domains (denoted by red nodes), two types of distribution function should be considered, including the inward distribution functions (as direction 3 in Region I) and the outward distribution functions (as direction 1 in Region I). As illustrated in Fig. 4 and Fig. 5(a), taking the D2Q9 lattice speed model as an example, the outward distribution functions for Region I, specifically f_1 , f_5 and f_8 at boundary node O , can be obtained by migrating the post-collision distribution functions from upstream nodes A_I , B_I , and C_I , as:

$$f_1(O) = f_1^*(O - c_1 \Delta t) = f_1^*(A_I) \quad (39)$$

$$f_8(O) = f_8^*(O - c_8 \Delta t) = f_8^*(B_I) \quad (40)$$

$$f_5(O) = f_5^*(O - c_5 \Delta t) = f_5^*(C_I) \quad (41)$$

The post-collision distribution function $f_1^*(A_I)$, $f_8^*(B_I)$ and $f_5^*(C_I)$ can be obtained via second-order interpolation according to Eqs. (36) and (37). Therefore, for Region I, the outward boundary distribution functions at the boundary can be calculated using the known information within the current region.

For the inward lattice direction, the distribution functions f_3 , f_6 , and f_7 must be calculated from their respective upstream post-collision distribution functions:

$$f_7(O) = f_7^*(O - c_7 \Delta t) = f_7^*(A_{II}) \quad (42)$$

$$f_3(O) = f_3^*(O - c_3 \Delta t) = f_3^*(B_{II}) \quad (43)$$

$$f_6(O) = f_6^*(O - c_6 \Delta t) = f_6^*(C_{II}) \quad (44)$$

However, these upstream nodes A_{II} , B_{II} , and C_{II} belong to Region II, which is external to Region I. To address, data must be transferred from Region II to Region I. Since the time steps are identical for both Regions I and II, the migration distances of f_3 , f_6 , and f_7 are also equivalent in these two regions, namely,

$$(c_3 \Delta t)_I = (c_3 \Delta t)_{II}, (c_6 \Delta t)_I = (c_6 \Delta t)_{II}, (c_7 \Delta t)_I = (c_7 \Delta t)_{II} \quad (45)$$

Therefore, the positions of A_{II} , B_{II} , and C_{II} in Region II are uniquely determined and satisfy the requirement of Eqs. (42)–(44).

The treatment of boundary distribution functions for Region II is similar to that of Region I with a notable exception: in Region II, f_3 , f_6 , and f_7 are classified as outward distribution functions, whereas f_1 , f_5 , and f_8 are classified as inward distribution functions. Furthermore, since f_2 and f_4 are parallel to the interface, their upstream values reside within the current region and can be directly migrated without requiring inter-mesh communication. In addition, as the migration speed of f_0 is 0, data exchange with other meshes is unnecessary.

By combining data from adjacent computational domains, the distribution functions can be accurately reconstructed at the intersection. From a computational perspective, each domain operates independently, utilizing unknown boundary distribution functions as boundary conditions and leveraging information provided by neighboring domains. This approach enables each computational domain to be self-contained, allowing for stable iterative calculations. Furthermore, since the interpolation relationship at the interface is solely dependent on grid conditions and remains invariant during calculations, the interpolation coefficients for the interface grid also can be pre-computed.

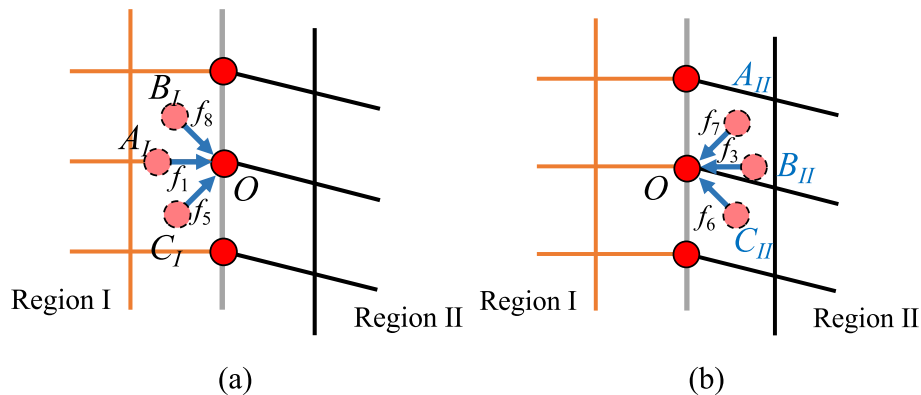


Fig. 5. Schematic of data transfer between different regions.

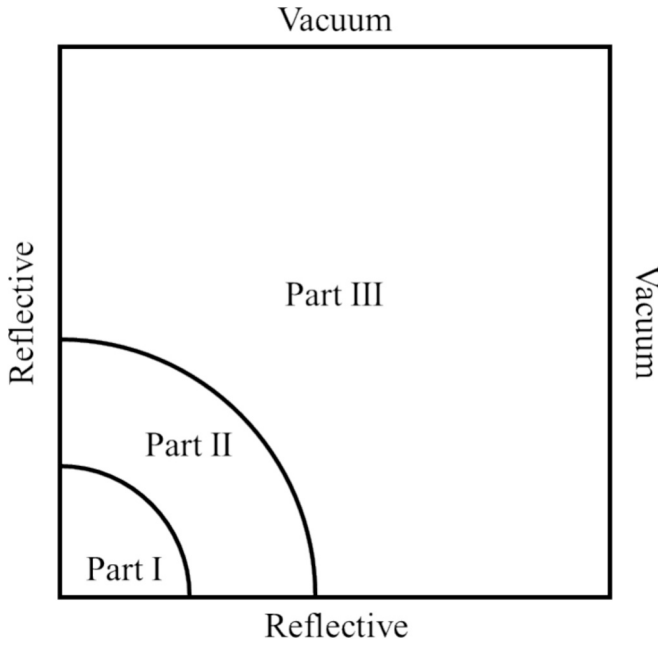


Fig. 6. Geometry of neutron shielding problem.

Table 1
Cross-sections of shielding problems.

	Σ_t (cm ⁻¹)	Σ_s (cm ⁻¹)	Q (cm ⁻³ s ⁻¹)
I	1.00	0.90	1.00
II	1.50	1.44	0.00
III	1.00	0.30	0.00

3. Results and discussions

This section presents simulations of three representative neutron transport problems to validate the accuracy and flexibility of the proposed GILBM.

3.1. Shielding problem

The first problem is a typical neutron shielding problem as referred to (Owens et al., 2017). As shown in Fig. 6, the shape of the central

source region is a quarter circle, surrounded by highly scattering material that is a quarter circle. Length of the square region is $L = 10$ cm, the radius of the first small circle (part I) is $r = 4/\sqrt{\pi}$ cm, and the radius of the second large circle (part II) is $R = 8/\sqrt{\pi}$ cm. Cross-sections of each region are shown in Table 1.

The GILBM is employed to simulate this shielding problem using S_6 angular discretization. To optimize the computational cost and accuracy, grid refinement is strategically applied in regions exhibiting significant gradient changes, whereas coarser meshes are utilized in areas with smooth variations. For comparative purposes, the SLBM is also applied to this problem using S_6 angular discretization. The mesh configurations for both GILBM and SLBM are illustrated in Fig. 7, in which the mesh numbers for GILBM and SLBM are 1904 and 22801, respectively. To demonstrate the configuration of the body-fitted mesh at the circular boundary, a localized enlargement of the mesh near the circular area is displayed in Fig. 8. This visualization demonstrates that the body-

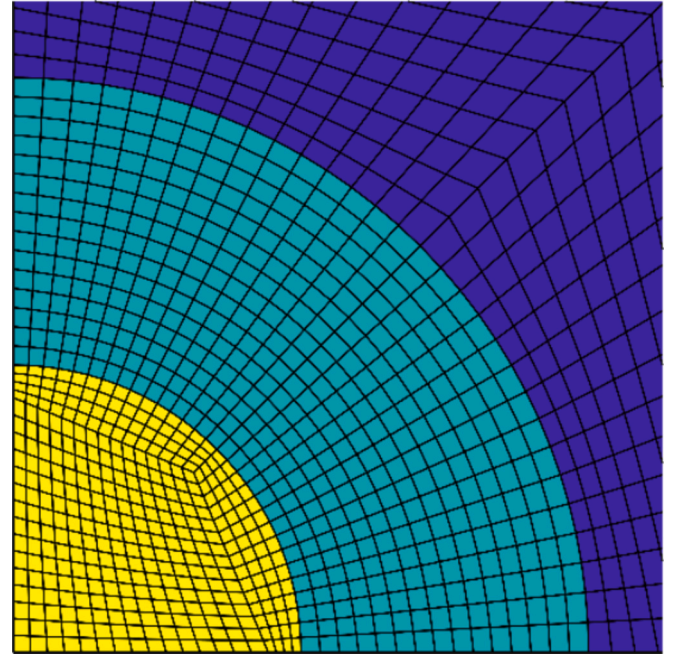


Fig. 8. Localized enlarged view of the body-fitted mesh.

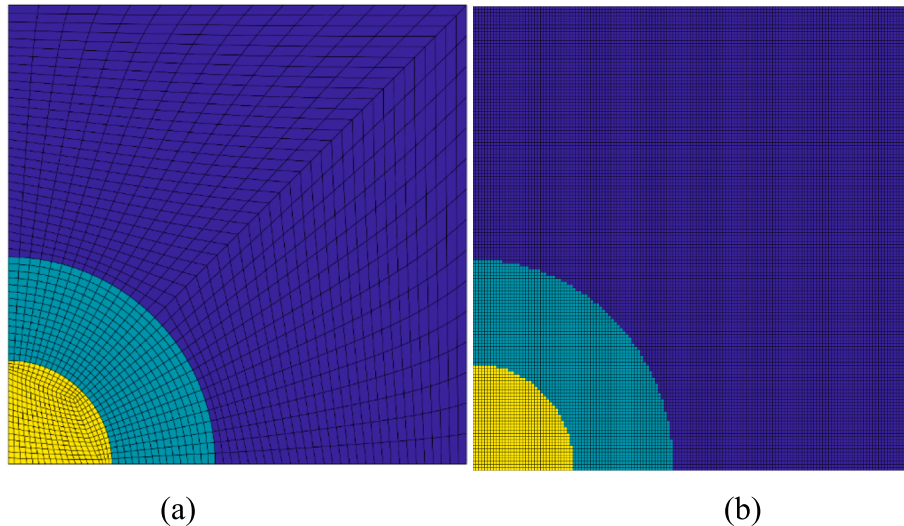


Fig. 7. Mesh configurations for the neutron shielding problem, including (a) GILBM meshes and (b) SLBM meshes.

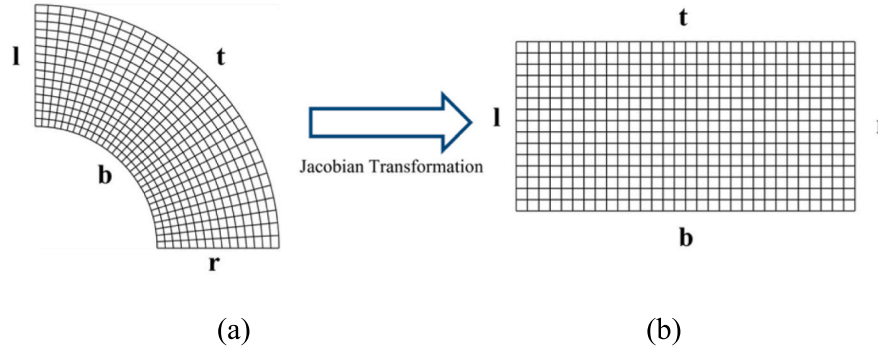


Fig. 9. Schematic of body-fitted mesh transformation of (a) irregular physical region and (b) regular computing region.

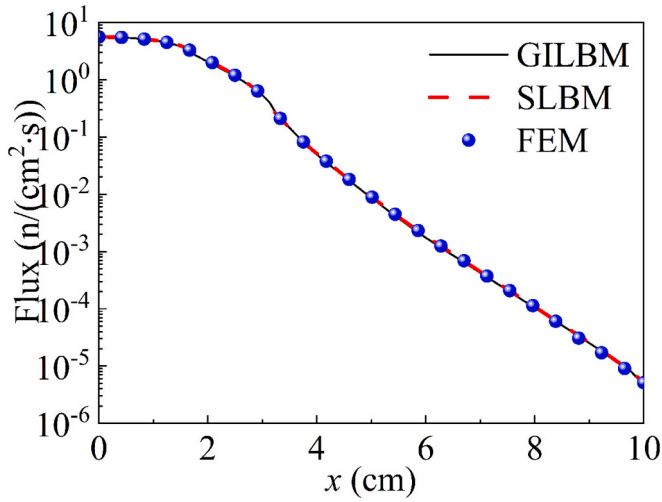


Fig. 10. Solutions of the neutron shielding problem.

Table 2
Comparisons between GILBM and SLBM.

	GILBM			SLBM		
Mesh number	1186	1636	1904	1936	10,201	22,801
CPU time (s)	163.61	204.73	311.12	10.85	165.55	771.55
Error (%)	0.4051	0.2606	0.0872	1.3678	0.2427	0.11

fitted meshes accurately capture the circular boundary, thereby effectively reducing the number of meshes required for complex geometries. Furthermore, Fig. 9 shows the transformation of body-fitted meshes, using Part II as a representative example. The mapping process involves transforming the irregular physical domain to a regular computational domain by aligning the four edges (l , r , b , t) of the physical domain with those of the computational domain. Subsequently, the structured meshes are divided into regular rectangular regions, as depicted in Fig. 9 (b). Through this mapping relationship, the coordinates of each mesh in the physical domain can be determined, as illustrated in Fig. 9(a).

Fig. 10 presents a comparative analysis of the neutron scalar flux along the diagonal line, as computed by GILBM and SLBM, alongside reference values obtained using the Finite Element Method (FEM) neutron transport solver DNTM (Wang et al., 2024) with S_6 angular discretization and 102,840 triangular meshes. Notably, the FEM solutions exhibit non-physical oscillations when coarser meshes are employed. In contrast, GILBM and SLBM demonstrate a lower mesh requirement for achieving high-accuracy solutions compared to FEM.

As evident from Fig. 10, the solutions obtained using the GILBM and the SLBM are in good agreement with those of the DNTM. The maximum

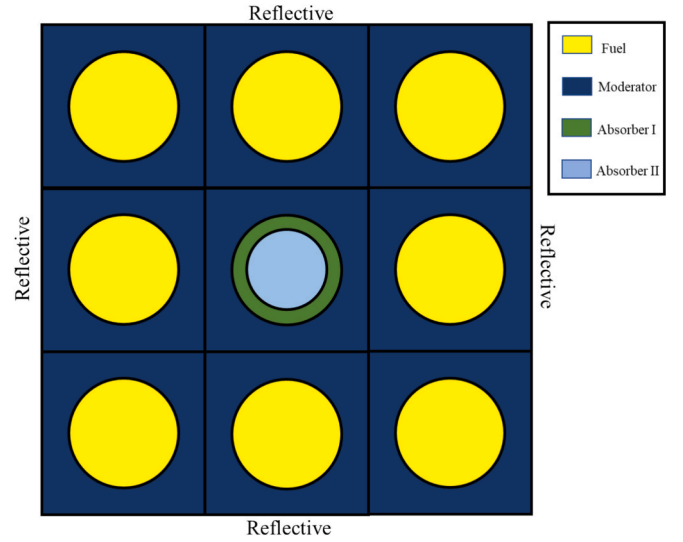


Fig. 11. Geometry of 3 × 3 fuel pin.

Table 3
Cross-sections of 3 × 3 fuel pin.

	Σ_t (cm ⁻¹)	Σ_s (cm ⁻¹)	Q (cm ⁻³ s ⁻¹)
Fuel	0.5145	0.3432	0.0
Moderator	0.7557	0.7504	1.0
Absorber I	21.63	0.3414	0.0
Absorber II	14.27	0.34	0.0

relative errors for GILBM and SLBM are 0.0872 % and 0.11 %, respectively, demonstrating their high accuracy. The presence of strong scattering region II and strong absorption region III materials outside the neutron source region I leads to a significant variation in neutron flux distribution along the diagonal, spanning seven orders of magnitude. Both SLBM and GILBM accurately capture the neutron distribution within the shielding material. Notably, when simulating curved boundary geometries, the SLBM requires refined meshes to ensure smoothness and achieve better results, whereas the GILBM can attain more accurate solutions with relatively coarser meshes. This advantage enables the GILBM to efficiently describe complex geometries using fewer computational resources compared to the SLBM.

To further assess the performance of the GILBM, a detailed comparison of its computational accuracy and CPU time with those of the SLBM is conducted under various mesh configurations. As presented in Table 2, the CPU time and maximum error of GILBM and SLBM are evaluated for different meshes, with relative errors calculated by comparing the results to DNTM solutions. Notably, to achieve a

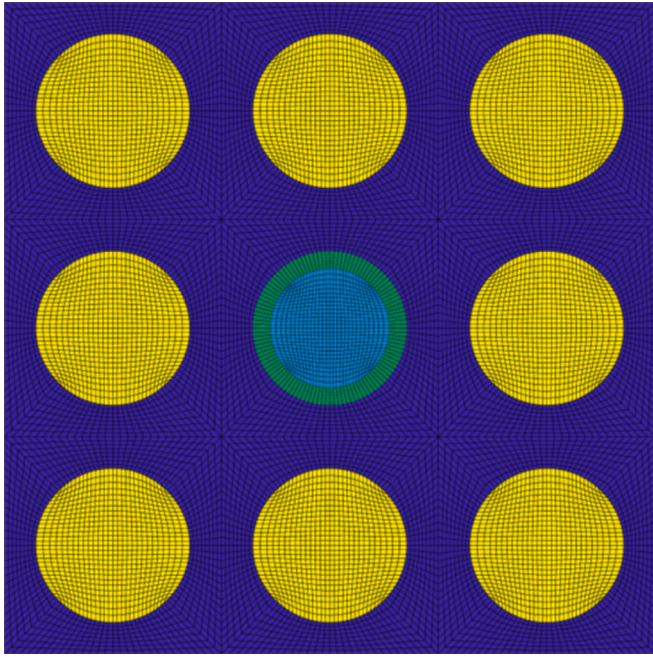


Fig. 12. Mesh configurations of 3×3 fuel pin.

calculation error of approximately 0.1 %, the SLBM requires 22,801 meshes and 771.55 s of computational time, whereas the GILBM achieves this level of accuracy with fewer than 1,900 meshes and less than 300 s of computation. Although the addition of interpolation steps increases the computational time of GILBM compared to SLBM at equivalent mesh densities, the significantly lower number of meshes required for GILBM to attain similar accuracy results in substantially reduced computational times. This advantage is expected to become more pronounced in simulations involving larger scales and more complex geometries.

3.2. 3×3 fuel pin

The second problem is a 3×3 fuel pin problem modified from (Wood and Williams, 1984). This problem is used to test the applicability of GILBM in dealing with complicated geometry with drastic changes in material properties. The geometry of this problem is shown in Fig. 11, in which an absorption pin is surrounded by eight fuel pins. All the physical boundaries are reflective. The length of each pin square is 2 cm. For the fuel pin, the radius of the fuel rod is 0.71 cm. For the absorption pin, the radii of absorbers I and II are 0.55 cm and 0.71 cm, respectively. Cross-sections of all these materials are given in Table 3, and the body-fitted mesh configuration of this problem is shown in Fig. 12.

The proposed GILBM serves as a detailed neutron transport solver. To facilitate a comparison of high-resolution neutron flux distributions for

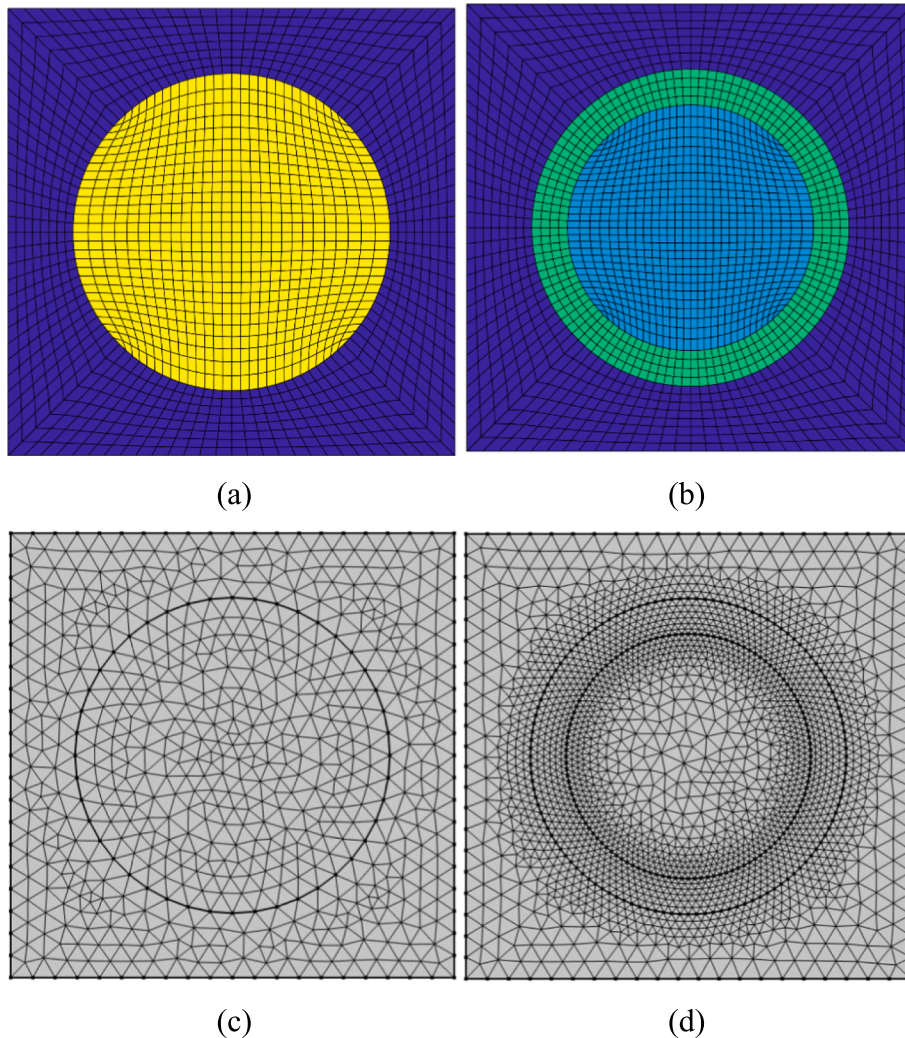


Fig. 13. Mesh configurations of (a) fuel pin and (b) absorption pin of GILBM and (c) fuel pin and (d) absorption pin of DNTM.

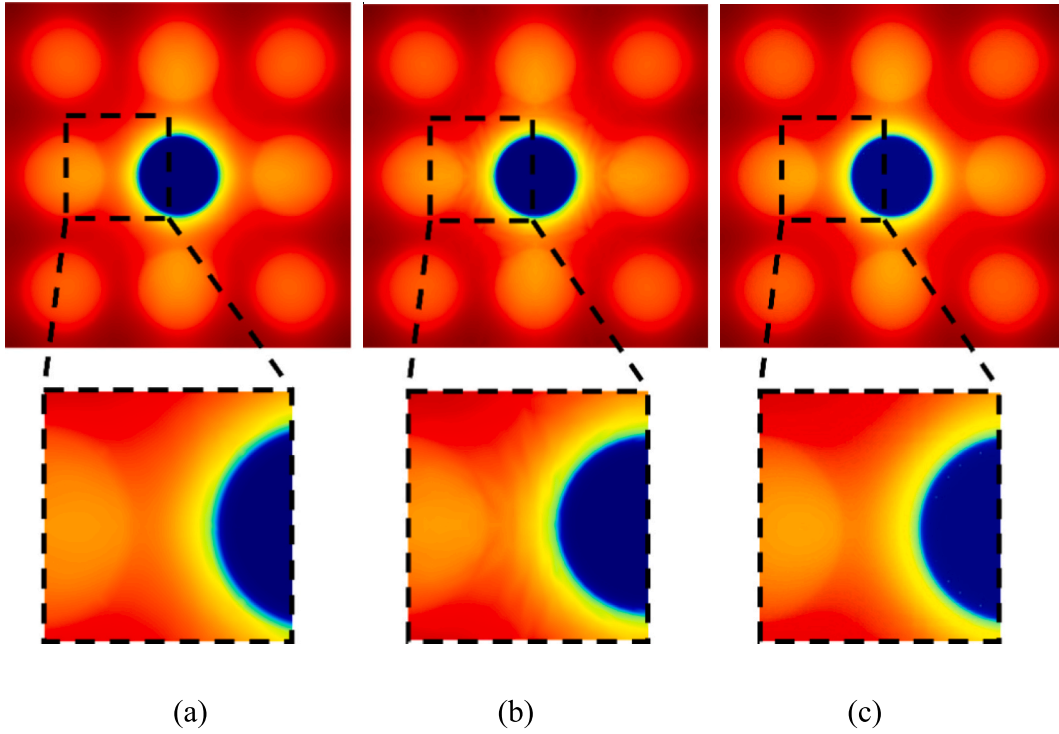


Fig. 14. Solutions of 3×3 fuel pin problem, including (a) GILBM solution, (b) FEM solution of DNTM and (c) referenced FEM solution (Owens et al., 2017).

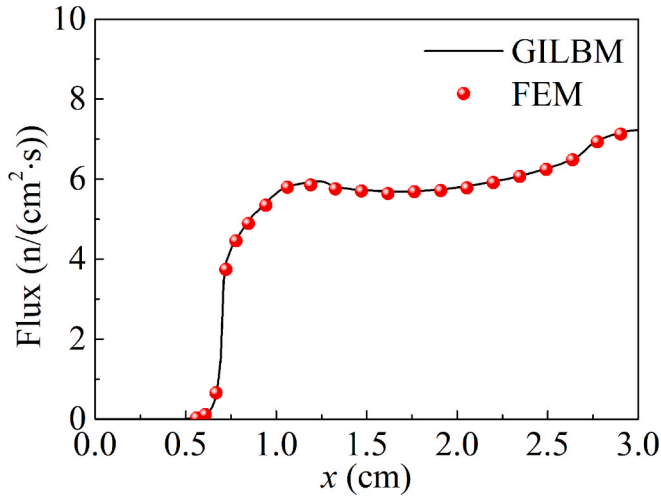


Fig. 15. Comparison of neutron scalar flux along the horizontal centerline.

this test case, a relatively dense mesh configuration is utilized. For clarity, the local meshes of fuel and absorption pins are magnified and illustrated in Fig. 13, including the GILBM meshes and FEM meshes. These figures demonstrate that the body-fitted meshes can effectively describe both fuel and absorption pins. Furthermore, the strong absorber material at the assembly's center is expected to cause significant variations in neutron flux. To accurately capture these fluctuations, a denser grid is employed. Specifically, this case utilizes 10,483 meshes with S_{24} angular discretization for GILBM.

Fig. 14(a) presents the neutron flux distribution calculated by GILBM, which is compared with two other solutions, including the FEM solution using DNTM (Wang et al., 2024) with S_{24} angular discretization and 13,112 meshes, as illustrated in Fig. 14(b), and the reference FEM solution (Owens et al., 2017) employing S_{32} angular discretization and 684,064 meshes, depicted in Fig. 14(c). A visual comparison reveals that

the proposed GILBM accurately simulates strongly non-uniform neutron transport problems, while requiring a significantly smaller number of meshes compared to both the DNTM and reference FEM solutions. The DNTM exhibits residual fluctuations in its calculation results due to inadequate mesh density between the absorber and fuel cells, as illustrated in the lower part of Fig. 14(b); in contrast, the GILBM effectively mitigates this issue.

Fig. 15 shows the neutron flux along the horizontal centerline, which is compared with the FEM solution of DNTM. Due to symmetry, only the results from the centroid to the boundary were compared. Because of the strong absorption characteristics of the central absorption pin, the neutron flux at this location is extremely low. Subsequently, due to the neutrons generated by the fuel pin, there is a large neutron gradient between the fuel and absorber pins. The maximum relative error of GILBM is 0.31244 %, indicating that for problems with sharp variations and complicated geometry, the GILBM can accurately calculate the flux distribution at various positions, and can still obtain relatively accurate results for areas with drastic changes.

3.3. Plate fuel problem

The last case is a plate fuel problem, whose geometric parameters are extracted from the JRR-3 M reactor (IWASAKI et al., 1984). The geometry of each fuel plate is shown in Fig. 16. The length and width of the structural material are 7.6 cm and 0.38 cm, respectively; the length and width of the moderator are 6.66 cm and 0.114 cm, respectively; the length and width of fuel are 6.16 cm and 0.076 cm, respectively. All the physical boundaries are reflective. Cross-sections of this problem are shown in Table 4, in which two energy groups are considered.

The characteristic of this type of problem is that the ratio of length to width is too large, but the variation of neutron flux in some areas is relatively gentle. For SLBM with uniform square meshes, a large number of meshes are required. In areas where the neutron flux variation is gentle, it is bound to waste computational resources due to the dense meshes. For GILBM, its interpolation process can ensure the collision and migration at each step, without being constrained by uniform square

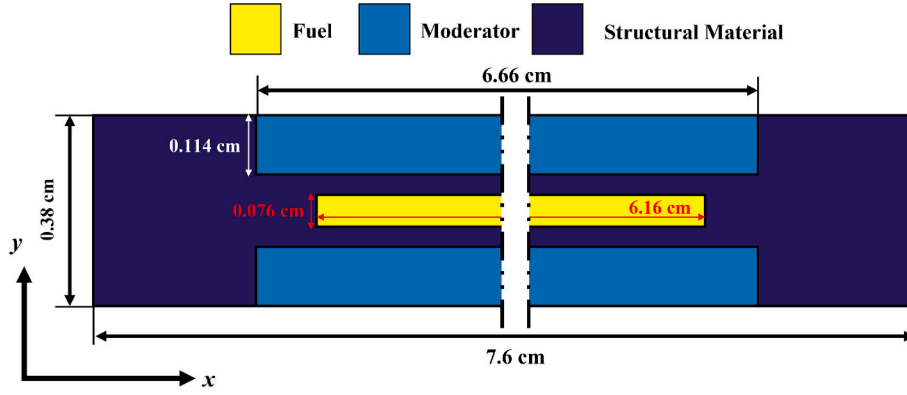


Fig. 16. Geometry of plate fuel problem.

Table 4
Cross-sections of plate fuel.

	Group	$\Sigma_{t,g}$	$\nu\Sigma_{f,g}$		$\Sigma_{s,g \rightarrow g'}$
Fuel	1	0.372634	0.0257031	1	0.341599
	2	0.77732	0.73501	2	0.000771592
Moderator	1	0.797233	0.0	1	0.000562749
				2	0.385537
	2	1.40232	0.0	1	0.566376
				2	0.2304213
Structural material	1	0.264868	0.0	1	0.000589448
				2	1.38085
	2	0.288459	0.0	1	0.263193
				2	0.000256652
				1	0.000490239
				2	0.282924

meshes. It allows users to create meshes more intelligently according to the problem situation, thereby saving computational resources.

This problem is simulated using S_6 angular discretization with GILBM and FEM. Half of the fuel pin is simulated by considering its symmetry. To accurately capture the spatial variations in neutron flux along the x -direction, a non-uniform mesh refinement strategy is employed, wherein a finer mesh is applied near the material boundaries and a coarser mesh is used towards the center. Ultimately, 200 non-uniform meshes are divided along the x -direction, while 200 uniformly distributed meshes are created along y -direction. The resulting mesh configuration is illustrated in Fig. 17, where the grid lines perpendicular to the x -axis have been omitted for the sake of clarity.

The normalized neutron fluxes for the fast and thermal groups along

the horizontal centerline are presented in Fig. 18, which are compared with those of DNTM. The distribution of fast neutrons, which are primarily generated through fission in the fuel region, exhibits a higher intensity within the fuel region than in the moderator region. In contrast, thermal neutrons, produced by the moderation of fast neutrons after they enter the moderator, display a higher distribution in the moderator region compared to the fuel region. Notably, both fast and thermal neutron groups exhibit a pronounced gradient near the fuel-moderator interface, owing to the abrupt change in material properties. This sharp gradient poses significant challenges for numerical simulation methods.

The eigenvalues calculated by GILBM and FEM are 1.52536 and 1.52492, respectively. The error of the eigenvalue is 44 pcm. For the normalized fast flux, the maximum error is located at $y = 3.5$ cm, which is 0.55347 %, while for the normalized thermal flux, the maximum error is located at $y = 0.0$ cm, which is 0.12471 %. These results indicate that the GILBM can maintain high accuracy when dealing with multi-group eigenvalue problems. For situations where there is significant difference in geometric aspect ratios or uneven distribution of neutron flux gradients, GILBM can save computational resources by using locally refined meshes.

4. Conclusions

The general interpolation-supplemented lattice Boltzmann method for the simulation of neutron transport in general body-fitted meshes was developed and implemented by adding an interpolation step in the standard neutron transport lattice Boltzmann model. The irregular physical domain is converted to a regular square computation domain

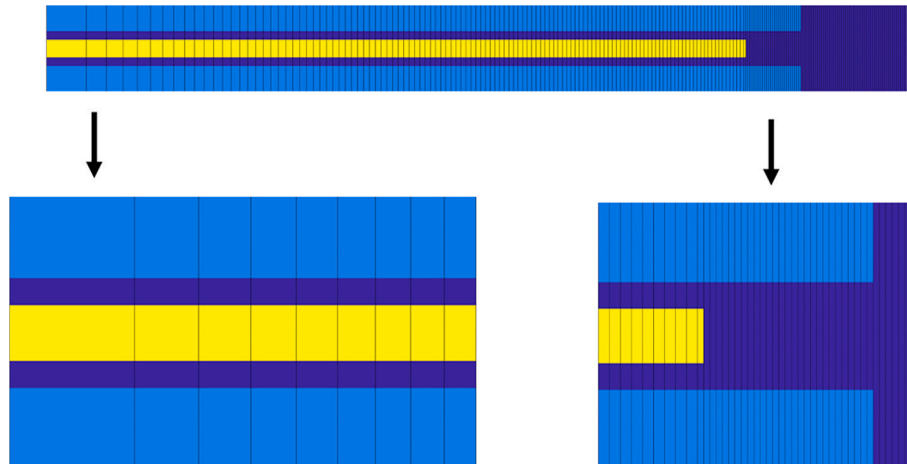


Fig. 17. Mesh configurations of the plate fuel problem.

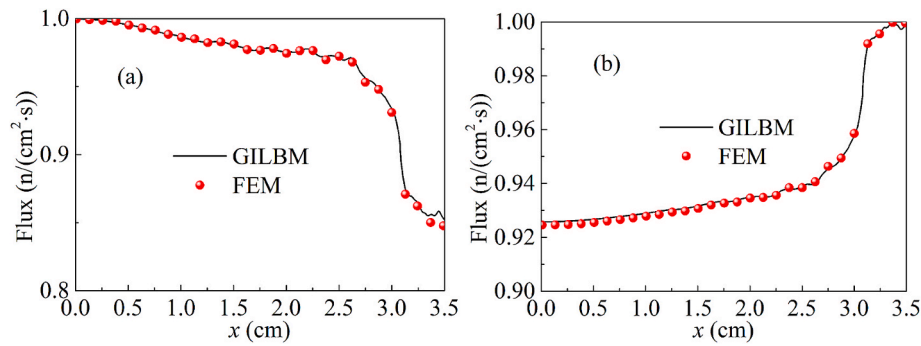


Fig. 18. Solutions of plate fuel problem, including (a) normalized fast group and (b) normalized thermal group.

by using the partial-differential equation method, and the lattice Boltzmann calculation is implemented in the computation domain. The only difference between the GILBM and SLBM is the addition of a new interpolation step between the migration and collision steps, which does not compromise the locality and strong parallelism of SLBM. Meanwhile, since the interpolation process is independent of the physical evolution, it can be precomputed before the calculation, thus the GILBM does not significantly increase the calculation time compared to SLBM. By simulating three typical benchmark problems, including the neutron shielding problem, the heterogeneous 3×3 fuel pin problem, and the plate fuel problem, the accuracy and flexibility of the proposed GILBM are verified. For further research, the proposed work can provide an optional technique for neutron transport simulation in complicated geometries.

It should be noted that the present study represents the first step in applying body-fitted LBM to reactor physics. The current verification is limited to the lattice and pin levels. Because the proposed GILBM is a detailed neutron transport solver, it requires a significant number of computational resources for whole-core large-scale simulations. Additionally, the interpolation-supplemented approach also increases the computational cost to some extent. In future research, to make this method more competitive and suitable for engineering applications, it is necessary to further simplify the implementation of the interpolation-supplemented process to improve efficiency. Moreover, this method should be extended to parallel computing domains, especially for GPU acceleration, taking full advantage of the strong locality of LBM to effectively enhance computational efficiency. Since all industrial and most academic neutron transport relies on either the integral or first-order transport equation, further in-depth comparisons with these methods, such as the method of characteristics or collision probabilities, are still needed in future works for engineering applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All necessary data has been added in manuscript.

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